GLA UNIVERSITY | DATA SCIENCE PROJECT

EMPLOYEE SALARY PREDICTION

A machine-learning study of employee and job attributes to estimate salary in USD.

TEAM 124_421

TEAM MEMBER	ROLL NUMBER
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PROJECT SNAPSHOT	**Domain:** Data Science / Machine Learning **Model:** Random Forest Regressor **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib
REPORT SCOPE	Notebook review, CSV profile, workflow, evaluation results, key observations, and practical improvements.

ACADEMIC PROJECT • SECOND YEAR

Prepared from the submitted notebook and employee_salary CSV dataset.

01 | Project Overview

The project builds a regression model that estimates annual employee salary in USD from role, experience, employment arrangement, work model, year, residence, company location, and company size.

OBJECTIVE	Explore salary patterns and train a model to predict salary_in_usd.
INPUTS	Eight features: job_title, experience_level, employment_type, work_models, work_year, employee_residence, company_location, company_size.
OUTPUT	Predicted salary in USD (salary_in_usd).
METHOD	One-hot encode categorical variables, split data 80/20, train Random Forest with 100 trees, then evaluate predictions.

02 | Dataset at a Glance

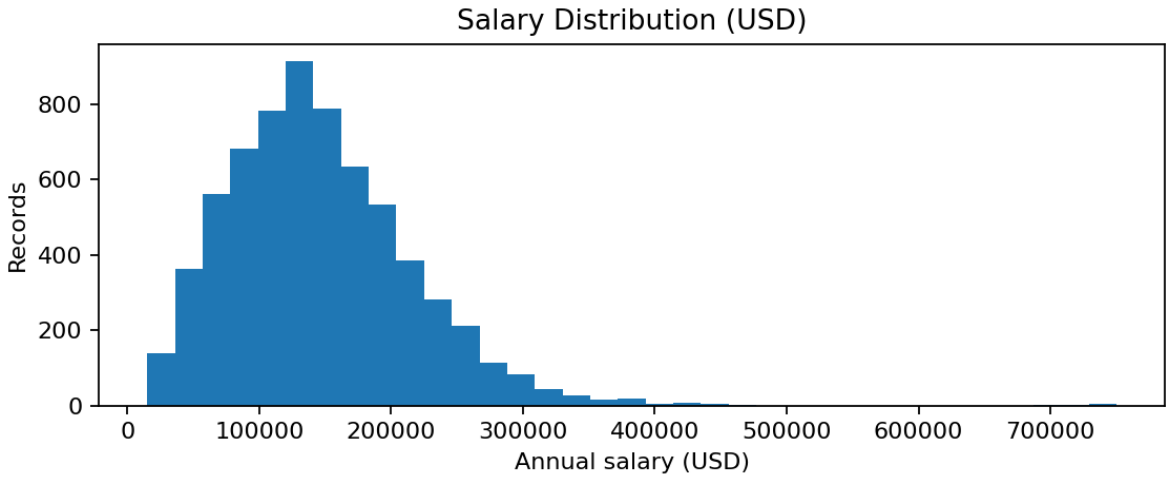
DATASET MEASURE	OBSERVED VALUE	NOTES
Rows in supplied CSV	11,087	Before cleaning
Columns	11	8 categorical/text + 3 integer columns
Duplicate rows	4,488	Removed by notebook cleaning step
Rows after duplicate removal	6,599	No missing target salary rows remained
Job titles / countries of residence	132 / 87	Distinct values
Work-year range	2020–2024	Years represented in the dataset

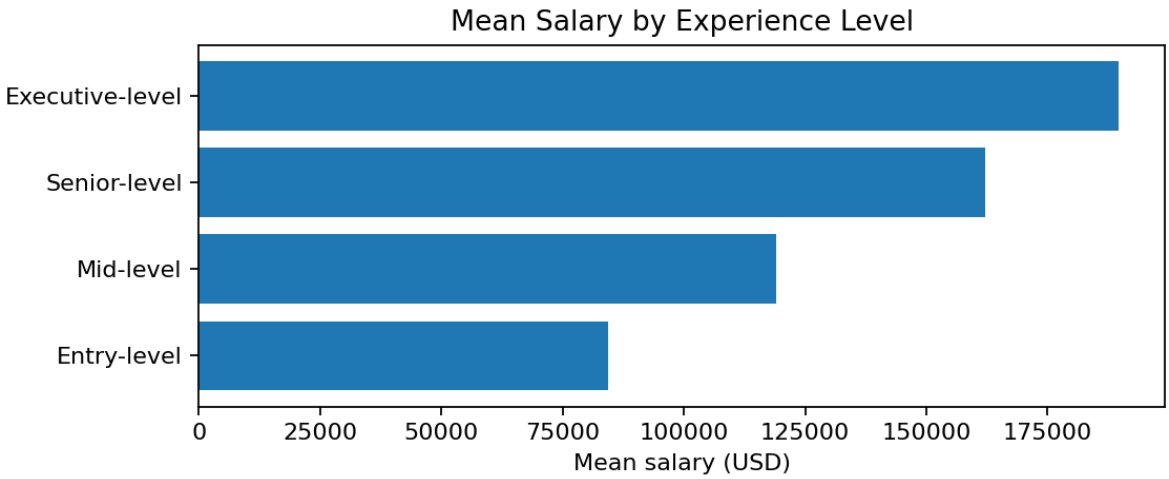
CSV COLUMN DICTIONARY

CSV FIELD	MEANING
job_title	Employee job title
experience_level	Entry, mid, senior or executive level
employment_type	Full-time, contract, part-time, freelance
work_models	On-site, remote or hybrid
work_year	Year of salary record
employee_residence	Employee's country of residence
salary	Original salary amount
salary_currency	Currency of original salary
salary_in_usd	Salary converted/recorded in USD; prediction target
company_location	Company country/location
company_size	Small, medium or large

03 | Analysis & Model Workflow

The notebook uses exploratory plots and grouped averages to inspect salary distribution, experience level, company size, popular job titles, and numeric correlations.





MODELLING PIPELINE

01 Load CSV	**02 Inspect & clean**	**03 Explore data**	**04 Select features**
05 One-hot encode	**06 80/20 split**	**07 Random Forest**	**08 Evaluate & save**

Cleaning in the notebook removes duplicate rows and rows with missing target salary. In the supplied CSV, all 11 columns are complete; the main reduction comes from duplicate removal.

04 | Model Results

The evaluation below was reproduced by running the notebook's stated Random Forest configuration on the supplied CSV after its cleaning steps, using an 80/20 split and random_state=42.

$44,664	$63,579	0.312
MAE · average absolute error	RMSE · error scale	R^2 · explained variation

METRIC	RESULT	INTERPRETATION
MAE	$44,663.63	Predictions differ from actual salaries by about this amount on average.
MSE	4,042,250,223 USD²	Squared error; more sensitive to large misses.
RMSE	$63,578.69	Typical error scale, expressed in USD.
R^2 score	0.3122	Moderate/limited fit; model explains about 31% of test-set variance.

05 | Key Findings

SALARY PROFILE	After duplicate removal, mean salary is **$145,561**; median is **$138,666**. Observed range: $15,000–$750,000.
DATA BALANCE	The data is concentrated in senior-level roles, full-time employment, US residence/company records, and medium-sized companies.
OVERALL ASSESSMENT	A solid end-to-end beginner/intermediate Data Science project: data cleaning, EDA, preprocessing, model training, evaluation, and model export are all represented.

06 | Improvements Before Submission

✓ Fix the new-employee example: the notebook uses labels such as “SE” and “FT”, but the CSV contains “Senior-level” and “Full-time”. Unknown labels are ignored by the encoder, which can weaken that example prediction.
✓ Investigate the 4,488 duplicate rows and document whether they are true duplicates or repeated observations.
✓ Compare against a baseline and alternative models; report cross-validation and feature importance.
✓ Add a scatter plot with a 45° reference line and explain limitations before using predictions for real pay decisions.

CONCLUSION	The project demonstrates a complete salary-prediction workflow. Current test performance ($R^2 \approx 0.31$) suggests the model captures some patterns but needs further tuning and validation before it can be considered highly accurate.
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